

第3章 拉格朗日力学 § 3.1 拉格朗日方程

1. 坐标变换关系、拉格朗日关系 ← 如何不推导而求得呢?

$$\hookrightarrow r_i = r_i(q_1, \dots, q_n, t) = r_i(q, t)$$

$$\dot{r}_i = \sum_{\alpha} \frac{\partial r_i}{\partial q_{\alpha}} \dot{q}_{\alpha} + \frac{\partial r_i}{\partial t}$$

$$\frac{\partial r_i}{\partial q_{\beta}} = \frac{\partial}{\partial q_{\beta}} \sum_{\alpha} \frac{\partial r_i}{\partial q_{\alpha}} \dot{q}_{\alpha} + \frac{\partial^2 r_i}{\partial q_{\beta} \partial t}$$

$$\frac{\partial r_i}{\partial t} = \frac{\partial r_i}{\partial t}(q, t) \text{ 与 } \dot{q} \text{ 无关} \quad \frac{\partial r_i}{\partial q_{\alpha}} \dot{q}_{\alpha} \text{ 与 } \dot{q}_{\beta} \text{ 无关 } (\alpha \neq \beta)$$

$$= \frac{\partial}{\partial q_{\beta}} \left(\frac{\partial r_i}{\partial q_{\alpha}} \dot{q}_{\alpha} \right) = \frac{\partial r_i}{\partial q_{\beta}} \Rightarrow \frac{\partial \dot{r}_i}{\partial \dot{q}_{\beta}} = \frac{\partial r_i}{\partial q_{\beta}}$$

$$\begin{aligned} \frac{\partial}{\partial q_{\beta}} r_i &= \frac{\partial}{\partial q_{\beta}} \sum_{\alpha} \frac{\partial r_i}{\partial q_{\alpha}} \dot{q}_{\alpha} + \frac{\partial r_i}{\partial q_{\beta} \partial t} \\ &= \frac{\partial}{\partial t} \left(\frac{\partial r_i}{\partial q_{\beta}} \right) + \sum_{\alpha} \frac{\partial}{\partial q_{\alpha}} \left(\frac{\partial r_i}{\partial q_{\beta}} \right) \dot{q}_{\alpha} = \frac{\partial}{\partial t} \left(\frac{\partial r_i}{\partial q_{\beta}} \right) \end{aligned}$$

$$\Rightarrow \frac{\partial}{\partial q_{\alpha}} \left(\frac{\partial}{\partial t} r_i \right) = \frac{\partial}{\partial t} \left(\frac{\partial}{\partial q_{\alpha}} r_i \right)$$

2. 拉格朗日方程 / 动力学方程

$$\sum_i (-m_i \ddot{r}_i) \cdot \delta \vec{r}_i = \sum_i -m_i \ddot{r}_i \cdot \left(\sum_{\alpha} \frac{\partial r_i}{\partial q_{\alpha}} \delta q_{\alpha} \right) = \sum_{\alpha} \sum_i \left[-m_i \right] \ddot{r}_i \cdot \frac{\partial r_i}{\partial q_{\alpha}} \delta q_{\alpha}$$

$$\text{其中 } \sum_i \ddot{r}_i \cdot \frac{\partial r_i}{\partial q_{\alpha}} = \sum_i \left[\frac{d}{dt} \left(\ddot{r}_i \cdot \frac{\partial r_i}{\partial \dot{q}_{\alpha}} \right) - \ddot{r}_i \cdot \frac{d}{dt} \left(\frac{\partial r_i}{\partial \dot{q}_{\alpha}} \right) \right]$$

$$= \sum_i \left[\frac{d}{dt} \left(\ddot{r}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{q}_{\alpha}} \right) - \ddot{r}_i \cdot \frac{\partial}{\partial q_{\alpha}} \left(\frac{\partial \vec{r}_i}{\partial t} \right) \right] = \sum_i \left[\frac{d}{dt} \left(\ddot{r}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{q}_{\alpha}} \right) - \ddot{r}_i \cdot \frac{\partial \ddot{r}_i}{\partial q_{\alpha}} \right]$$

$$= \sum_i \left[\frac{d}{dt} \frac{\partial}{\partial \dot{q}_{\alpha}} \left(\frac{1}{2} \ddot{r}_i^2 \right) - \frac{\partial}{\partial q_{\alpha}} \left(\frac{1}{2} \ddot{r}_i^2 \right) \right] \quad \text{利用1.中关系}$$

$$\Rightarrow \sum_i (\ddot{r}_i - m_i \ddot{r}_i) \cdot \delta \vec{r}_i = \sum_{\alpha} \left(Q_{\alpha} - \sum_i m_i \left[\frac{d}{dt} \frac{\partial}{\partial \dot{q}_{\alpha}} \left(\frac{1}{2} \ddot{r}_i^2 \right) - \frac{\partial}{\partial q_{\alpha}} \left(\frac{1}{2} \ddot{r}_i^2 \right) \right] \right) \delta q_{\alpha}$$

$$= \sum_{\alpha} \left[Q_{\alpha} - \frac{d}{dt} \frac{\partial T}{\partial \dot{q}_{\alpha}} + \frac{\partial T}{\partial q_{\alpha}} \right] \cdot \delta q_{\alpha} = 0 \quad \text{其中 } T = \sum_i \frac{1}{2} m_i \ddot{r}_i^2$$

$$\Rightarrow Q_{\alpha} = \frac{d}{dt} \frac{\partial T}{\partial \dot{q}_{\alpha}} - \frac{\partial T}{\partial q_{\alpha}} \quad \text{对完整系统的拉格朗日方程}$$

* 那完整系统 (有那完整约束) 呢?

假设有 k 个约束 $f_L(q, \dot{q}, t) = 0$ 线性的速度约束

$$= \sum_{\alpha} A_{L\alpha}(q) dq_{\alpha} + A_L dt$$

等时变分: $\sum_{\alpha} A_{L\alpha}(q) \delta q_{\alpha} = 0$

达朗贝尔: $(\vec{F} - m\vec{\ddot{r}}) \cdot \delta \vec{r} = 0$

$$\hookrightarrow \sum_{\alpha} \left(\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_{\alpha}} - \frac{\partial T}{\partial q_{\alpha}} + A_{L\alpha} \right) \delta q_{\alpha} = 0$$

拉格朗日乘子法:

$$\sum_{\alpha} \left(A_{L\alpha} + \frac{d}{dt} \frac{\partial T}{\partial \dot{q}_{\alpha}} - \frac{\partial T}{\partial q_{\alpha}} + \lambda_L A_{L\alpha} \right) \delta q_{\alpha} = 0$$

可这样取 k 个 λ_L 使 $\left(A_{L\alpha} + \frac{d}{dt} \frac{\partial T}{\partial \dot{q}_{\alpha}} - \frac{\partial T}{\partial q_{\alpha}} + \lambda_L A_{L\alpha} \right) = 0$

剩下 $s-1$ 个 δq_{α} 独立,

$\Rightarrow$

第二种的欧拉 拉格朗日方程

$$F = F(x, y, y')$$

$$\text{若 } F \text{ 满足 } \frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0 \quad \frac{\partial F}{\partial x} + \frac{d}{dx} \left(F - y' \frac{\partial F}{\partial y'} \right) = 0$$

$$\frac{dF}{dx} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} y' + \frac{\partial F}{\partial y'} y''$$

$$= \frac{\partial F}{\partial y} + \frac{d}{dx} \left(y' \frac{\partial F}{\partial y'} \right) \left(\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} \right) y'$$

$$\Rightarrow \frac{\partial F}{\partial y} + \frac{d}{dx} \left(y' \frac{\partial F}{\partial y'} - F \right) = 0$$

若 F 不显含 x , $\frac{\partial F}{\partial x} = 0$

$$\hookrightarrow y' \frac{\partial F}{\partial y'} - F = C \quad \text{对应一个守恒量}$$

例: 推导平面极坐标中质点的运动方程

$$\vec{r} = \rho \cdot \hat{\rho} \quad T = \frac{1}{2} m [\dot{\rho}^2 + (\rho \dot{\varphi})^2]$$

$$Q_\rho = \frac{d}{dt} \frac{\partial T}{\partial \dot{\rho}} - \frac{\partial T}{\partial \rho} = \frac{d}{dt} (m \dot{\rho}) - m \rho \dot{\varphi}^2 = m \ddot{\rho} - m \dot{\varphi}^2 \rho$$

$$Q_\varphi = \frac{d}{dt} \frac{\partial T}{\partial \dot{\varphi}} - \frac{\partial T}{\partial \varphi} = \frac{d}{dt} (m \rho^2 \dot{\varphi}) = 2m \rho \dot{\varphi} \dot{\rho} + m \rho^2 \ddot{\varphi}$$

$$Q_\rho = \sum_i \vec{F}_i \cdot \frac{\partial \vec{r}_i}{\partial \rho} = \vec{F} \cdot \hat{\rho} = F_\rho \quad Q_\varphi = \vec{F} \cdot \frac{\partial \vec{r}}{\partial \varphi} = \vec{F} \cdot \rho \hat{\varphi} = \rho F_\varphi$$

$$F_\rho = m \ddot{\rho} - m \dot{\varphi}^2 \rho$$

$$F_\varphi = 2m \dot{\varphi} \dot{\rho} + m \rho \ddot{\varphi}$$

3. 主动力是保守力时

$$\vec{F}_i = -\nabla_i V$$

$$Q_\alpha = \sum_i \vec{F}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{q}_\alpha} = \sum_i -\nabla_i V \cdot \frac{\partial \vec{r}_i}{\partial \dot{q}_\alpha} = -\frac{\partial V}{\partial \dot{q}_\alpha}$$

$$\rightarrow -\frac{\partial V}{\partial \dot{q}_\alpha} = \frac{d}{dt} \frac{\partial T}{\partial \dot{q}_\alpha} - \frac{\partial T}{\partial \dot{q}_\alpha} \quad \text{令 } L = T - V, \text{ 为拉格朗日函数}$$

$$\rightarrow \frac{\partial L}{\partial \dot{q}_\alpha} = \frac{d}{dt} \frac{\partial T}{\partial \dot{q}_\alpha} \quad \text{又因 } V \text{ 仅是 } q, t \text{ 的函数} \quad \frac{\partial T}{\partial \dot{q}_\alpha} = \frac{\partial L}{\partial \dot{q}_\alpha}$$

$$\rightarrow \frac{\partial L}{\partial \dot{q}_\alpha} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_\alpha} = 0$$

$$\text{定义广义初量 } p_\alpha = \frac{\partial L}{\partial \dot{q}_\alpha} = \frac{\partial T}{\partial \dot{q}_\alpha} \quad (\text{当 } V \text{ 与 } \dot{q}_\alpha \text{ 无关}), \text{ 拉格朗日力 } R_\alpha = \frac{\partial L}{\partial q_\alpha}$$

$$\rightarrow \text{拉氏方程: } \dot{p}_\alpha = Q_\alpha + R_\alpha$$

(可证运动守恒)

$$\text{eg. 平动中 } \vec{r}(q, q_\alpha + dq_\alpha, t) = \vec{r}(q, q_\alpha, t) + dq_\alpha \vec{n}$$

$$p_\alpha = \frac{\partial L}{\partial \dot{q}_\alpha} = \frac{\partial T}{\partial \dot{q}_\alpha} = \frac{d}{dq_\alpha} \left(\frac{1}{2} m \vec{r}^2 \right) = m \vec{r} \cdot \frac{\partial \vec{r}}{\partial \dot{q}_\alpha}$$

$$= (m \dot{\vec{r}}) \cdot \frac{\partial \vec{r}}{\partial q_a} = m \dot{\vec{r}} \cdot \vec{n}$$

$$Q_a = \vec{F} \cdot \frac{\partial \vec{r}}{\partial q_a} = \vec{F} \cdot \vec{n}$$

转动中 $\vec{r}(q, q_a + dq_a, t) = \vec{r}(q, q_a, t) + \vec{n} \times \vec{r}$ ($\vec{n}$ 是转轴方向)

$$p_a = m \dot{\vec{r}} \cdot \frac{\partial \vec{r}}{\partial q_a} = m \dot{\vec{r}} \cdot (\vec{n} \times \vec{r}) = m \vec{n} \cdot (\vec{r} \times \dot{\vec{r}}) = \vec{n} \cdot [\vec{r} \times (m \dot{\vec{r}})]$$

$$= \vec{n} \cdot \vec{L}$$

$$Q_a = \vec{F} \cdot \frac{\partial \vec{r}}{\partial q_a} = \vec{F} \cdot (\vec{n} \times \vec{r}) = \vec{n} \cdot (\vec{r} \times \vec{F}) = \vec{n} \cdot \vec{M}$$

1. 拉格朗日函数并不唯一

(1) 不同 γ 型符号选取

(2) 同一组 γ 型符号

$$(i) L' = CL$$

$$(ii) L' = L + \frac{d}{dt} \phi(q, t)$$

$$\frac{d}{dt} \phi(q, t) = \sum_a \frac{\partial \phi}{\partial q_a} \dot{q}_a + \frac{\partial \phi}{\partial t}$$

$$\frac{d}{dt} \left(\frac{\partial}{\partial \dot{q}_a} \left(\frac{d\phi}{dt} \right) \right) = \frac{d}{dt} \left(\frac{\partial \phi}{\partial \dot{q}_a} \right) = \frac{\partial}{\partial q_a} \left(\frac{d\phi}{dt} \right)$$

$$\hookrightarrow \frac{\partial L'}{\partial q_a} - \frac{d}{dt} \frac{\partial L'}{\partial \dot{q}_a} = \frac{\partial L}{\partial q_a} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_a} + \frac{\partial}{\partial q_a} \left(\frac{d\phi}{dt} \right) - \frac{d}{dt} \left[\frac{\partial}{\partial \dot{q}_a} \left(\frac{d\phi}{dt} \right) \right]$$

$$= \frac{\partial L}{\partial q_a} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_a} = 0$$

L' 与 L 满足同样精确的拉格朗日方程

2. 对于 s 个 γ 型符号, 有 s 个拉格朗日方程

但 γ 型符号与 γ 速度独立, 还有 s 个动力学方程

$$\dot{q}_a = \frac{d}{dt} \langle q_a \rangle$$

3. 势函数与 $\vec{v} \times$ 速度有关时, 仍可能有拉格朗日方程

例: 证明摆线 $\begin{cases} x = a(\theta + \sin\theta) & \text{或等时性} \\ y = a(1 - \cos\theta) \end{cases}$

§ 3.2 运动积分 诺特定理

1. 可速坐标 广义动量积分

如 $\frac{\partial L}{\partial q_0} = 0$, 称 q_0 为可速坐标

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_0} \right) = \frac{d}{dt} (P_0) = 0 \quad \text{即 } P_0 = \text{const.}$$

例: m, M, θ 已知, 求小楔子下滑加速度

$$T = \frac{1}{2} M \dot{X}^2 + \frac{1}{2} m [\dot{q} \cos \theta + \dot{X}]^2 + (\dot{q} \sin \theta)^2$$

$$V = mgq \sin \theta$$

$$L = T - V$$



$$X: \frac{\partial L}{\partial X} = 0 \quad \text{故 } P_X = \frac{\partial L}{\partial \dot{X}} = [M\dot{X} + m(\dot{X} + \dot{q} \cos \theta)] = 0 \quad (X \text{ 是可速坐标})$$

$$\Rightarrow \dot{q} = -\frac{m+M}{m \cos \theta} \dot{X}, \quad \ddot{q} = -\frac{m+M}{m \cos \theta} \ddot{X}$$

$$q: \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \frac{d}{dt} [m(\dot{q} + \dot{X} \cos \theta)] + mg \sin \theta$$

$$= m(\ddot{q} + \ddot{X} \cos \theta) + mg \sin \theta$$

$$\Rightarrow \ddot{X} = \frac{mg \sin \theta}{\frac{m+M}{m \cos \theta} - \cos \theta} = \frac{mg \sin \theta \cos \theta}{m + M \sin^2 \theta}$$

$$\ddot{q} = -\frac{(m+M)g \sin \theta}{m + M \sin^2 \theta}$$

2. 广义能量积分.

$$L = L(q, \dot{q}, t) \quad \dot{L} = \sum_j \left(\frac{\partial L}{\partial \dot{q}_j} \dot{q}_j + \frac{\partial L}{\partial q_j} q_j \right) + \frac{\partial L}{\partial t}$$

拉格朗日方程: $\frac{\partial L}{\partial q_\alpha} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_\alpha} = 0$

$$\begin{aligned} \dot{L} &= \frac{\partial L}{\partial t} + \sum_\alpha \left[\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_\alpha} \dot{q}_\alpha + \frac{\partial L}{\partial q_\alpha} \dot{q}_\alpha \right] \\ &= \frac{\partial L}{\partial t} + \frac{d}{dt} \left(\sum_\alpha \frac{\partial L}{\partial \dot{q}_\alpha} \dot{q}_\alpha \right) = \frac{\partial L}{\partial t} + \frac{d}{dt} \sum_\alpha \underline{p}_\alpha \cdot \dot{q}_\alpha \end{aligned}$$

$$\text{即: } \frac{d}{dt} \left(\sum_\alpha \underline{p}_\alpha \cdot \dot{q}_\alpha - L \right) + \frac{\partial L}{\partial t} = 0$$

定义 $H = \sum_\alpha \underline{p}_\alpha \cdot \dot{q}_\alpha - L$ 为广义能量函数

$$\Rightarrow H + \frac{\partial L}{\partial t} = 0$$

当 $L = L(q, \dot{q})$ $H = \text{const.}$ 为广义能量积分 / 雅可比积分

① 当约束是定常的 / $r_i = r_i(q, \dot{q})$

$$\underline{r}_i = \sum_\alpha \frac{\partial r_i}{\partial \dot{q}_\alpha} \dot{q}_\alpha$$

$$\begin{aligned} T &= \sum_i \frac{1}{2} m_i \underline{\dot{r}}_i \cdot \underline{\dot{r}}_i = \sum_i \frac{1}{2} m_i \sum_\alpha \frac{\partial r_i}{\partial \dot{q}_\alpha} \dot{q}_\alpha \cdot \sum_\beta \frac{\partial r_i}{\partial \dot{q}_\beta} \dot{q}_\beta \\ &= \sum_i \sum_\alpha \sum_\beta \frac{1}{2} m_i \frac{\partial r_i}{\partial \dot{q}_\alpha} \frac{\partial r_i}{\partial \dot{q}_\beta} \dot{q}_\alpha \dot{q}_\beta \end{aligned}$$

$$\begin{aligned} \frac{\partial T}{\partial \dot{q}_\alpha} &= \sum_i \frac{1}{2} m_i \left(\frac{\partial r_i}{\partial \dot{q}_\alpha} \sum_\beta \frac{\partial r_i}{\partial \dot{q}_\beta} \dot{q}_\beta + \frac{\partial r_i}{\partial \dot{q}_\alpha} \sum_\beta \frac{\partial r_i}{\partial \dot{q}_\beta} \dot{q}_\beta \right) \\ &= \frac{\partial r_i}{\partial \dot{q}_\alpha} \sum_i \sum_\beta m_i \frac{\partial r_i}{\partial \dot{q}_\beta} \dot{q}_\beta \end{aligned}$$

$$H = \sum_\alpha \frac{\partial L}{\partial \dot{q}_\alpha} \dot{q}_\alpha - L = \sum_\alpha \frac{\partial T}{\partial \dot{q}_\alpha} \dot{q}_\alpha - L = 2T - L = T + V$$

此时广义能量为机械能

② 当约束是恒定的 / $r_i = r_i(q, t)$

$$\dot{r}_i = \frac{\partial r_i}{\partial t} + \sum_a \frac{\partial r_i}{\partial q_a} \dot{q}_a$$

$$T = \sum_i \frac{1}{2} m_i \dot{r}_i \cdot \dot{r}_i = \sum_i \frac{1}{2} m_i \left(\frac{\partial r_i}{\partial t} + \sum_a \frac{\partial r_i}{\partial q_a} \dot{q}_a \right) \cdot \left(\frac{\partial r_i}{\partial t} + \sum_a \frac{\partial r_i}{\partial q_a} \dot{q}_a \right)$$

$$= \sum_i \frac{1}{2} m_i \left[\sum_a \sum_b \frac{\partial r_i}{\partial q_a} \cdot \frac{\partial r_i}{\partial q_b} \dot{q}_a \dot{q}_b + 2 \sum_a \frac{\partial r_i}{\partial q_a} \dot{q}_a \cdot \frac{\partial r_i}{\partial t} + \left(\frac{\partial r_i}{\partial t} \right)^2 \right]$$

$$= \underbrace{\sum_i \sum_a a_{00a} \dot{q}_a \dot{q}_a}_{T_2} + \underbrace{\sum_a a_{1a} \dot{q}_a}_{T_1} + \underbrace{a_0}_{T_0}$$

$$\sum_a \frac{\partial L}{\partial \dot{q}_a} \dot{q}_a = \sum_a \frac{\partial T}{\partial \dot{q}_a} \dot{q}_a = 2T_2 + T_1 + 0T_0 = 2T_2 + T_1$$

$$H = \sum_a \frac{\partial T}{\partial \dot{q}_a} \cdot \dot{q}_a - L = T_2 - T_0 + V \quad \text{变换非定常时, } H \text{ 不等于机械能.}$$

例: 单摆.

$$V = -mgl \cos \theta \quad T = \frac{1}{2} m (l \dot{\theta})^2$$

$$L = T - V = \frac{1}{2} m (l \dot{\theta})^2 + mgl \cos \theta$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = l \ddot{\theta} + g \sin \theta = 0$$

$$\frac{\partial L}{\partial t} = 0$$

$$\hookrightarrow H = \sum p \dot{q} - L = \frac{\partial L}{\partial \dot{\theta}} \cdot \dot{\theta} - L = \frac{1}{2} m l \dot{\theta}^2 - mgl \cos \theta$$

$$= \frac{1}{2} m (l \dot{\theta})^2 - mgl \cos \theta$$

$$\dot{\theta} = \sqrt{\frac{2g}{l} (\cos \theta - \cos \theta_0)} + \dot{\theta}_0^2$$

$$\hookrightarrow \ddot{\theta} = \frac{d\dot{\theta}}{dt} = \frac{d\dot{\theta}}{d\theta} \dot{\theta}$$

$$l \dot{\theta} \ddot{\theta} + g \sin \theta = 0 \quad \Rightarrow \quad \ddot{\theta} d\theta = -\frac{g}{l} \sin \theta d\theta$$

$$\dot{\theta} = \sqrt{\frac{2g}{l}} (\cos \theta - \cos \theta_0) + \theta_0^2 = \sqrt{\frac{2g}{l}} |\cos \theta - \cos \theta_0| + \frac{l}{2g} \theta_0^2$$

$$\textcircled{1} \frac{1}{2g} \theta_0^2 - \cos \theta_0 \in [-1, 1)$$

此时有最大摆角 α

$$\cos \alpha = \cos \theta_0 - \frac{l}{2g} \theta_0^2$$

$$\Rightarrow \frac{d\theta}{|\cos \theta - \cos \alpha|} = \sqrt{\frac{2g}{l}} dt$$

$$\int \cos = (1 - 2 \sin^2 \frac{\theta}{2})$$

$$\frac{d\theta}{|2| \sin^2 \frac{\theta}{2} - \sin^2 \frac{\theta_0}{2}} = \sqrt{\frac{2g}{l}} dt$$

$$\text{若代 } \sin \frac{\theta}{2} = \sin \frac{\theta_0}{2} \sin \varphi \quad \frac{1}{2} \cos \frac{\theta}{2} d\theta = \sin \frac{\theta_0}{2} \cos \varphi d\varphi$$

$$\frac{2 \sin(\frac{\theta_0}{2}) \cos \varphi}{\cos \frac{\theta_0}{2}} d\varphi$$

$$\sin(\frac{\theta_0}{2}) \cos \varphi$$

*

$$\vec{r}_i \rightarrow \vec{r}_i' = \vec{r}_i + \delta \vec{r}_i \quad L = (q, \dot{q})$$

$$\delta L = \sum_i \frac{\partial L}{\partial q_i} \delta q_i + \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i$$

$$= \sum_i \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \delta q_i + \frac{\partial L}{\partial \dot{q}_i} \frac{d}{dt} (\delta q_i)$$

$$= \frac{d}{dt} \sum_i \left(\frac{\partial L}{\partial \dot{q}_i} \delta q_i \right)$$

$$= \frac{d}{dt} \left(\sum_i \frac{\partial T}{\partial \vec{r}_i} \frac{\partial \vec{r}_i}{\partial \dot{q}_i} \delta q_i \right)$$

$$= \frac{d}{dt} \sum_i (m_i \dot{\vec{r}}_i \cdot \delta \vec{r}_i)$$

3. 诺特定理

在时间 t 或者广义坐标 q_i 的一系列无穷小变换下,

如果系统的作用量 S 不变, 则一定有一个不变的运动积分

无穷小变换: $\begin{cases} \bar{t} = t + \sum_i \varepsilon_i \zeta_i(q, t) & \text{为什么是函数! (而不是 } t+t) \\ \bar{q}_\alpha = q_\alpha + \sum_i \varepsilon_i \eta_{\alpha i}(q, t) & L = L(q, \dot{q}, t) \text{ 是 } t \text{ 的函数} \end{cases}$

作用量 $S = \int_{t_1}^{t_2} L dt$ $\hookrightarrow$ 无穷小变换为函数

$$\bar{S} = \int_{\bar{t}_1}^{\bar{t}_2} \bar{L}(\bar{q}, \dot{\bar{q}}, \bar{t}) d\bar{t} \quad t > \bar{t} \quad q \rightarrow \bar{q}$$

$$= \int_{t_1}^{t_2} \bar{L}(\bar{q}, \dot{\bar{q}}, \bar{t}) \frac{d\bar{t}}{dt} dt \Rightarrow \bar{L}(\bar{q}, \dot{\bar{q}}, \bar{t}) \frac{d\bar{t}}{dt} = L(q, \dot{q}, t)$$

若系统的力学性质在无穷小变换下不变 $\bar{L}$

则 $\bar{L}(q, \dot{q}, t)$ 与 $L(q, \dot{q}, t)$ 满足同一形式的 $E-L$ 方程

$$\hookrightarrow \bar{L}(\bar{q}, \dot{\bar{q}}, \bar{t}) = L(\bar{q}, \dot{\bar{q}}, \bar{t}) + \frac{d\Lambda(\bar{q}, \bar{t})}{d\bar{t}} \quad \text{称为形式协变性}$$

(形式不变性: $\bar{L}(\bar{q}, \dot{\bar{q}}, \bar{t}) = L(q, \dot{q}, t)$)

$$L(q, \dot{q}, t) dt = [L(\bar{q}, \dot{\bar{q}}, \bar{t}) + \frac{d}{d\bar{t}} \Lambda(\bar{q}, \bar{t})] d\bar{t}$$

$$\hookrightarrow L(q, \dot{q}, t) = \frac{d\bar{t}}{dt} \left[L(\bar{q}, \dot{\bar{q}}, \bar{t}) + \frac{d}{d\bar{t}} \Lambda(\bar{q}, \bar{t}) \right]$$

分别处理各部份

$$\frac{d\bar{t}}{dt} = 1 + \sum_i \varepsilon_i \frac{d\zeta_i}{dt}$$

$$\dot{\bar{q}}_\alpha = \frac{d\bar{q}_\alpha}{d\bar{t}} = \frac{dt}{d\bar{t}} \frac{d}{dt} \left[q_\alpha + \sum_i \varepsilon_i \eta_{\alpha i}(q, t) \right]$$

$$= \frac{1}{1 + \sum_i \varepsilon_i \frac{d\zeta_i}{dt}} \left(\dot{q}_\alpha + \sum_i \varepsilon_i \frac{d\eta_{\alpha i}}{dt} \right)$$

$$\approx \left(\dot{q}_\alpha + \sum_i \varepsilon_i \frac{d\eta_{\alpha i}}{dt} \right) \left(1 - \sum_i \varepsilon_i \frac{d\zeta_i}{dt} \right)$$

$$\approx \dot{q}_\alpha + \sum_i \varepsilon_i \left(\frac{d\eta_i}{dt} - \dot{q}_\alpha \frac{dz_i}{dt} \right)$$

将 $L(\bar{q}, \dot{\bar{q}}, \bar{t})$ 展开

$$\begin{aligned} L(\bar{q}, \dot{\bar{q}}, \bar{t}) &= L(q, \dot{q}, t) + \sum_{\alpha=1}^2 \left[\frac{\partial L}{\partial q_\alpha} (\bar{q}_\alpha - q_\alpha) + \frac{\partial L}{\partial \dot{q}_\alpha} (\dot{\bar{q}}_\alpha - \dot{q}_\alpha) \right] + \frac{\partial L}{\partial t} (\bar{t} - t) \\ &= L(q, \dot{q}, t) + \sum_{\alpha=1}^2 \left[\frac{\partial L}{\partial q_\alpha} \cdot \sum_i \varepsilon_i \eta_i(q, t) + \frac{\partial L}{\partial \dot{q}_\alpha} \sum_i \varepsilon_i \left(\frac{d\eta_i}{dt} - \dot{q}_\alpha \frac{dz_i}{dt} \right) \right] \\ &\quad + \frac{\partial L}{\partial t} \sum_i \varepsilon_i \zeta_i(q, t) \\ &= L(q, \dot{q}, t) + \sum_i \sum_{\alpha} \left[\frac{\partial L}{\partial q_\alpha} \varepsilon_i \eta_i(q, t) + \frac{\partial L}{\partial \dot{q}_\alpha} \varepsilon_i \left(\frac{d\eta_i}{dt} - \dot{q}_\alpha \frac{dz_i}{dt} \right) \right] \\ &\quad + \frac{\partial L}{\partial t} \sum_i \varepsilon_i \zeta_i(q, t) \end{aligned}$$

$$\frac{d}{dt} L(\bar{q}, \bar{t}) = \frac{d}{dt} \left[\underline{L(q, t)} + \frac{\partial L}{\partial q} (\bar{q} - q) + \frac{\partial L}{\partial t} (\bar{t} - t) \right]$$

所以通过 $= \frac{d}{dt} \left[\underline{L(q, t)} + \frac{\partial L}{\partial q} \sum_i \varepsilon_i \eta_i(q, t) + \frac{\partial L}{\partial t} \sum_i \varepsilon_i \zeta_i(q, t) \right]$

规范选取 $= \frac{d}{dt} \left[\underline{L(q, t)} + \frac{\partial L}{\partial q} \sum_i \varepsilon_i \left[\frac{\partial L}{\partial q} \eta_i(q, t) + \frac{\partial L}{\partial t} \zeta_i(q, t) \right] \right]$

为0?

$$L(q, \dot{q}, t) = L(\bar{q}, \dot{\bar{q}}, \bar{t}) \frac{dt}{d\bar{t}} + \frac{d}{dt} L(\bar{q}, \bar{t})$$

$$= (1 + \sum_i \varepsilon_i \frac{dz_i}{dt})$$

$$\{ L(q, \dot{q}, t) + \sum_i \sum_{\alpha} \left[\frac{\partial L}{\partial q_\alpha} \varepsilon_i \eta_i(q, t) + \frac{\partial L}{\partial \dot{q}_\alpha} \varepsilon_i \left(\frac{d\eta_i}{dt} - \dot{q}_\alpha \frac{dz_i}{dt} \right) \right] + \frac{\partial L}{\partial t} \sum_i \varepsilon_i \zeta_i(q, t) \}$$

$$+ \frac{d}{dt} \left[\underline{L(q, t)} + \frac{\partial L}{\partial q} \sum_i \varepsilon_i \left[\frac{\partial L}{\partial q} \eta_i(q, t) + \frac{\partial L}{\partial t} \zeta_i(q, t) \right] \right] \quad \text{令其等于0}$$

$$= L(q, \dot{q}, t) \sum_i \varepsilon_i \frac{dz_i}{dt} + \frac{d}{dt} \left[\underline{L(q, t)} + \frac{\partial L}{\partial q} \sum_i \varepsilon_i \left[\frac{\partial L}{\partial q} \eta_i(q, t) + \frac{\partial L}{\partial t} \zeta_i(q, t) \right] \right]$$

$$L(q, \dot{q}, t) + \sum_i \sum_{\alpha} \left[\frac{\partial L}{\partial q_\alpha} \varepsilon_i \eta_i(q, t) + \frac{\partial L}{\partial \dot{q}_\alpha} \varepsilon_i \left(\frac{d\eta_i}{dt} - \dot{q}_\alpha \frac{dz_i}{dt} \right) \right] + \frac{\partial L}{\partial t} \sum_i \varepsilon_i \zeta_i(q, t)$$

$$\hookrightarrow \sum_i \varepsilon_i \left\{ \underline{L(q, \dot{q}, t) \frac{dz_i}{dt}} + \frac{d}{dt} \left[\sum_{\alpha} \frac{\partial L}{\partial q_\alpha} \eta_{\alpha i}(q, t) + \frac{\partial L}{\partial t} \zeta_i(q, t) \right] + \frac{\partial L}{\partial t} \zeta_i(q, t) \right\}$$

$$+ \sum_i \left[\frac{\partial L}{\partial q_\alpha} \eta_i(q, t) + \frac{\partial L}{\partial \dot{q}_\alpha} \left(\frac{d\eta_i}{dt} - \dot{q}_\alpha \frac{dz_i}{dt} \right) \right] = 0 \quad 1+2 = \frac{d}{dt} (L \sum_i \varepsilon_i) - \sum_i \left(\frac{\partial L}{\partial q_\alpha} \dot{q}_\alpha + \frac{\partial L}{\partial \dot{q}_\alpha} \dot{q}_\alpha \right)$$

$$\Rightarrow \frac{d}{dt} \left[\sum_{\alpha} \frac{\partial L}{\partial q_\alpha} \eta_{\alpha i} + \frac{\partial L}{\partial t} \zeta_i + L \sum_i \varepsilon_i + \sum_{\alpha} \frac{\partial L}{\partial \dot{q}_\alpha} (\eta_i - \dot{q}_\alpha \zeta_i) \right] \quad 3 = \frac{d}{dt} \left[\sum_{\alpha} \frac{\partial L}{\partial q_\alpha} (\eta_i - \dot{q}_\alpha \zeta_i) \right]$$

$$= \sum_i \frac{\partial L}{\partial \dot{q}_\alpha} \ddot{z}_i q_i + \frac{\partial L}{\partial \dot{q}_\alpha} \ddot{z}_i \dot{q}_\alpha + \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_\alpha} \eta_i - q_\alpha \ddot{z}_i > \\ - \frac{\partial L}{\partial \dot{q}_\alpha} \ddot{z}_i \dot{q}_\alpha - \frac{\partial L}{\partial \dot{q}_\alpha} \eta_i = \left(\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_\alpha} - \frac{\partial L}{\partial \dot{q}_\alpha} \right) (\eta_i - q_\alpha \ddot{z}_i) = 0$$

$$\Rightarrow \text{存在运动积分 } I_i(q, t) = \sum_\alpha \frac{\partial L}{\partial \dot{q}_\alpha} \eta_{\alpha i} + \frac{\partial L}{\partial t} \ddot{z}_i + L \ddot{z}_i + \sum_i \frac{\partial L}{\partial \dot{q}_\alpha} (\eta_i - q_\alpha \ddot{z}_i) \\ H = \sum p_\alpha \dot{q}_\alpha - L \\ = \sum (\eta_{\alpha i} p_\alpha - H \ddot{z}_i + \frac{\partial L}{\partial \dot{q}_\alpha} \eta_{\alpha i} + \frac{\partial L}{\partial t} \ddot{z}_i)$$

$$\text{例: } L = L(q, \dot{q}), \quad \eta_i = 0$$

$$1) \text{ 作时间平移变换, 取 } \ddot{z}_i = 1, \quad \eta_{\alpha i} = 0$$

$$\text{运动积分 } I_i = -H = \text{const.} \rightarrow \text{能量守恒}$$

$$2) \text{ 作空间平移变换, 取 } \ddot{z}_i = 0, \quad \eta_{\alpha i} = 1 \quad (\text{对 } i = 1, 2, \dots)$$

$$I = \sum \eta_{\alpha i} p_\alpha = p_\alpha \rightarrow \text{动量守恒} \quad (\text{在 } \alpha \text{ 方向平移} \rightarrow \alpha \text{ 方向动量守恒})$$

§ 3.3 非完整系统的动力学

广义坐标下的虚功原理给出:

$$\sum_{\alpha} (Q_{\alpha} - \frac{d}{dt} \frac{\partial T}{\partial \dot{q}_{\alpha}} + \frac{\partial T}{\partial q_{\alpha}}) \delta q_{\alpha} = 0$$

而 q_{α} 不独立, 受非完整约束: $f_r(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t) = 0 \quad (r=1, \dots, k)$

线性非完整约束: $\sum_{\alpha} A_{r\alpha}(q, t) \dot{q}_{\alpha} + A_r(q, t) = 0$

$$\text{乘 } dt \rightarrow \sum_{\alpha} A_{r\alpha}(q, t) dq_{\alpha} + A_r(q, t) dt = 0$$

$$\text{当 } dt=0, \quad dq_{\alpha} \rightarrow \delta q_{\alpha}$$

$$\rightarrow \sum_{\alpha} A_{r\alpha}(q, t) \delta q_{\alpha} = 0$$

$$\text{一般地, } \sum_{\alpha} \frac{\partial f_r}{\partial \dot{q}_{\alpha}} \delta q_{\alpha} = 0 \quad \text{称为切塔耶夫条件}$$

与拉氏方程进行 L 求教法:

$$\sum_{\alpha} (Q_{\alpha} - \frac{d}{dt} \frac{\partial T}{\partial \dot{q}_{\alpha}} + \frac{\partial T}{\partial q_{\alpha}} + \lambda_r \frac{\partial f_r}{\partial \dot{q}_{\alpha}}) \delta q_{\alpha} = 0 \quad \alpha=1, \dots, n, \quad r=1, \dots, k$$

选取 k 个 $\lambda_r \rightarrow () = 0 \rightarrow$ 剩余 δq_{α} 独立

$\rightarrow$ 全部 $() = 0$

$$\text{即: } \frac{d}{dt} \frac{\partial T}{\partial \dot{q}_{\alpha}} - \frac{\partial T}{\partial q_{\alpha}} = Q_{\alpha} + \sum_r \lambda_r \frac{\partial f_r}{\partial \dot{q}_{\alpha}} \quad (4)$$

+ k 个非完整约束 f_r

$\Rightarrow$ 解出 s 个 q_{α} 和 k 个 λ

如主动力保守: $Q_\alpha = -\frac{\partial V}{\partial q_\alpha}$

$$\rightarrow \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_\alpha} - \frac{\partial L}{\partial q_\alpha} = \sum_r \lambda_r \frac{\partial f_r}{\partial q_\alpha} \quad (\text{非完整约束力全量})$$

§ 3.4 推广

保守系中, $Q_\alpha = -\frac{\partial V}{\partial q_\alpha}$ 又有 $\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_\alpha} - \frac{\partial T}{\partial q_\alpha} = Q_\alpha$

定义 $L = T - V$, 则 $\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_\alpha} - \frac{\partial L}{\partial q_\alpha} = 0$

这是因为 $\frac{\partial V}{\partial \dot{q}_\alpha} = 0$, 称 V 为普通势

若广义力 $Q_\alpha = \frac{d}{dt} \frac{\partial U}{\partial \dot{q}_\alpha} - \frac{\partial U}{\partial q_\alpha}$ 定义 $L = T - U$

$$\rightarrow \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_\alpha} - \frac{\partial L}{\partial q_\alpha} = 0$$

为使 $\ddot{q}_\alpha$ 不出现, U 关于 $\dot{q}_\alpha$ 线性

即 $U = \sum_\alpha U_\alpha(q, t) \dot{q}_\alpha + U_0(q, t)$ 称为广义势

eg. $\vec{F} = e(\vec{v} \times \vec{B} + \vec{E})$

其中 $\vec{v} \times \vec{B} \sim \frac{d}{dt} \frac{\partial U}{\partial \dot{q}_\alpha}$, $\vec{E} \sim -\frac{\partial U}{\partial q_\alpha}$

$$\hookrightarrow U = e(\varphi - \vec{v} \times \vec{A})$$